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- (54) **STRAWBERRY PLANT NAMED ‘UC KEYSTONE’**
- (50) Latin Name: *Fragaria x ananassa* Duchesne
Varietal Denomination: **UC Keystone**
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- (52) **U.S. Cl.**
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CPC *A01H 6/7409* (2018.05)

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See application file for complete search history.

(56) **References Cited**

PUBLICATIONS

See information in accompanying Information Disclosure Statement letter.
Cal Poly Strawberry Center 2022 Strawberry Field Day, retrieved online at https://content-calpoly-edu.s3.amazonaws.com/strawberry/1/images/2022FieldDayBooklet-FINAL-07_22_22-PM-comp.pdf, 6 introductory pages, pp. 20-29. (Year: 2022).
U.S. Appl. No. 18/135,705, Non-Final Office Action dated Aug. 29, 2023, 9 pages.

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(57) **ABSTRACT**

‘UC Keystone’ is a ‘day-neutral’ flowering cultivar of a strawberry plant that produces exceptionally high yields of firm, long shelf-life fruit and is resistant to *Fusarium* wilt, *Verticillium* wilt, and *Phytophthora* crown rot.

2 Drawing Sheets

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Genus and species: The strawberry plant of this invention is botanically known as *Fragaria x ananassa* Duchesne.

Variety denomination: The variety denomination is ‘UC Keystone’.

BACKGROUND

This invention relates to a new and distinct day-neutral strawberry cultivar designated as ‘UC Keystone’, which originated from a cross performed in the winter of 2017. The plant of this selection was originally designated ‘17C138P062’.

BRIEF SUMMARY

‘UC Keystone’ is a *Fusarium* wilt resistant day-neutral cultivar that originated in the winter of 2017 from a cross between ‘UCD Moxie’ (U.S. Plant Pat. No. 32,953) and ‘UCD Warrior’ (U.S. Plant Pat. No. 32,950). It is not known which is the male parent and which is the female parent. Clones of ‘UC Keystone’ were initially propagated from a single mother plant in 2018. ‘UC Keystone’ has since been preserved by annual cycles of asexual propagation from stolons in a facility at Winters, California.

‘UC Keystone’ was selected from a full-sib family (17C138) generated from the cross between ‘UCD Moxie’ and ‘UCD Warrior’. ‘UC Keystone’ shares one of the same parents, ‘UCD Moxie’, with ‘UC Golden Gate’ (U.S. Plant patent application Ser. No. 18/135,710, filed of even date). Seeds of the 17C138 family and the full-sib family 17C139, from which ‘UC Golden Gate’ originated, were harvested

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from greenhouse-grown plants in the spring of 2017 and germinated in June 2017. Seedlings were transplanted to a greenhouse in July 2017, hardened off in a shade house in August 2017, and transplanted to the field in September 2017. ‘UC Keystone’ was one of 10,000 individual hybrid plants from 397 full-sib families that were grown in 2017-18). The population was visually phenotyped in the spring of 2018 to: (a) identify individuals with outstanding fruit size, firmness, symmetry, color, gloss, and visual appeal; (b) eliminate individuals with fruit defects and deformities; (c) estimate marketable fruit yields; (d) identify putative photoperiod insensitive individuals; (e) assess stolon proliferation (runner production) and plant architecture; and (f) select individuals for clonal multiplication and further testing. Ninety-four individuals from the 2017-18 selection cycle were selected for on-farm advanced testing (Phase I, II, and III) in day-neutral production environments, from Nipomo in southern California to Prunedale in northern California. Disease resistance of the selections was tested in Davis, California.

‘UC Keystone’ differs from the closest comparison varieties as follows: Parent ‘UCD Moxie’ has a “reduced runnering” phenotype and produces fewer runners than ‘UC Keystone’. ‘UC Keystone’ has improved tolerance to *Macrophomina* compared to ‘UCD Moxie’, which is susceptible. ‘UC Keystone’ also has moderate resistance to *Verticillium* wilt compared to parent ‘UCD Warrior’, which is moderately susceptible. ‘UC Keystone’ is less compact and has shorter flower trusses than related variety ‘UC Golden Gate’ and produces higher yields of fruit in the late season compared to ‘UC Golden Gate’, which produces more fruit

early in the season. Further comparison varieties ‘Monterey’ (U.S. Plant Pat. No. 19,767), ‘UCD Royal Royce’ (U.S. Plant Pat. No. 32,952) and ‘UCD Valiant’ (U.S. Plant Pat. No. 32,984) employed in comparative field trials described herein are susceptible to Fusarium wilt, in contrast to ‘UC

Keystone’. ‘UC Keystone’ was genotyped with a 50,000-SNP array (Hardigan et. al., *Frontiers in Plant Science* 10:1789, 2020; Hardigan et al., *Mol. Biol. Evol.* 38:2285-2305, 2021) that included 72 cultivars owned by the University of California and 300 publicly available cultivars not owned by the University of California. After quality- and LD-pruning, 31,212 SNP markers with well-separated codominant genotypic clusters were selected for further analysis. These analyses confirmed that ‘UC Keystone’ is genetically distinct from its parent varieties, related variety ‘UC Golden Gate’, and comparison varieties ‘Monterey’, ‘UCD Royal Royce’, and ‘UCD Valiant’, as well as all of the other cultivars evaluated.

BRIEF DESCRIPTION OF THE DRAWINGS

The colors in the photograph are depicted as nearly true as is reasonably possible to obtain in color reproductions of this type.

FIG. 1 depicts fruit of ‘UC Keystone’.
FIG. 2 shows 5-month old ‘UC Keystone’ plants.

DETAILED DESCRIPTION

Botanical Description

Botanical descriptors of ‘UC Keystone’ are provided in Table 1. The descriptors were collected in the spring and summer of 2022 from plants grown in Santa Maria, California. Plants evaluated in the spring were 6 months old. Plants evaluated in the summer were 8 months old. Colors are designated with reference to The Royal Horticultural Society (R.H.S.) Colour Chart, Sixth Edition, 2015. The characteristics of ‘UC Keystone’ may vary in detail, depending upon environmental factors and culture conditions.

TABLE 1

“Mean values were estimated from nine samples per trait.			
Category	Trait	Unit	Range (Mean ^a) or Color (RHS Colour Designation)
Foliar	Plant height	mm	300-360(328)
Foliar	Plant spread	mm	520-620(566)
Foliar	Plant growth habit		Semi-upright
Foliar	Position of the inflorescence in relation to foliage		Above
Foliar	Mid-tier leaflet length	mm	90-120(102)
Foliar	Mid-tier leaflet width	mm	110-140(129)
Foliar	Petiole length	mm	230-290(250)
Foliar	Stipule core color	color code	Moderate Yellow Green (146 C)
Foliar	Stipule margin color	color code	Moderate Yellow Green (146 A)
Foliar	Stolons/nursery mother plant	count	10-12(11.0)
Foliar	Stolon color	color code	Brilliant Yellow Green (150 C)
Foliar	Leaf color, adaxial	color code	Moderate Olive Green (147 A)

TABLE 1-continued

“Mean values were estimated from nine samples per trait.			
Category	Trait	Unit	Range (Mean ^a) or Color (RHS Colour Designation)
Foliar	Leaf color, abaxial	color code	Moderate Yellow Green (147 B)
Foliar	Leaf, blistering		Medium
Foliar	Leaf, glossiness		Medium
Foliar	Leaf, shape of the base of terminal leaflet		Obtuse
Flower	Petal, arrangement of petals		Overlapping
Flower	Petal number	count	5-6(5)
Flower	Petal length	mm	10-15(12)
Flower	Petal width	mm	10-15(12)
Flower	Calyx diameter	mm	25-40(35)
Flower	Corolla diameter	mm	19-27(21)
Flower	Calyx, size in relation to corolla		Larger
Flower	Flower, Stamen		Present
Flower	Sepal number	count	10-15(11)
Flower	Calyx Color	color code	Moderate Olive Green (137 A)
Flower	Upper Petal Color	color code	White (NN155 D)
Flower	Lower Petal Color	color code	White (NN155 D)
Flower	Stamen number	count	22-32(27)
Fruit	Achene Color	color code	Greenish Brilliant Yellow (4 A)
Fruit	Position of achenes		Below Surface
Fruit	Fruit length	mm	43-56(50)
Fruit	Fruit width	mm	30-38(34)
Fruit	Shape		Conical
Fruit	Size of hollow core	mm	0-10(6)
Fruit	External Fruit (exterior) color	color code	Vivid Red (44 A)
Fruit	Fruit flesh color	color code	Vivid Reddish Orange (33 A)
Fruit	Fruit core color	color code	Moderate Reddish Orange (35 A&B)
Fruit	Fruit size	g/fruit	22-30(26)

The time of the beginning of flowering is late February, 4 months post-planting.

Fusarium Resistance

Marker-assisted selection was employed to identify individuals predicted to be heterozygous or homozygous for the dominant (favorable) alleles for a Fusarium wilt resistance locus (FW1) and the PERPETUAL FLOWERING (PF) locus. The genotypes predicted by PF- and FW-linked SNPs for ‘UC Keystone’ were PFpf and FW1fw1. The predicted genotypes were confirmed with a 50,000 Aximo SNP array (Hardigan et al, 2020).

The dominant FW1 allele confers resistance to *Fusarium* wilt race 1, cause by *Fusarium oxysporum* f.sp. *fragariae*. *Fusarium* resistance was further tested by three years of replicated testing of bare-root plants artificially inoculated with AMP132, a race 1 isolate of the pathogen. The resistance of ‘UC Keystone’ to *Fusarium* wilt was compared to that of ‘UC Golden Gate’, ‘San Andreas’ (U.S. Plant Pat. No. 19,975), which is heterozygous FW1fw1 *Fusarium* wilt-resistant; and to susceptible (homozygous recessive (fw1fw1)) varieties ‘Monterey’, ‘UCD Royal Royce’, and ‘UCD Valiant’. ‘UC Keystone’, ‘UC Golden Gate’, and ‘San Andreas’ were symptomless in these *Fusarium* wilt screening trials over three years (Table 2) with the mean disease scores in the highly resistant range and were not significantly different. The mean disease scores for ‘Monterey’,

‘UCD Royal Royce’, and ‘UCD Valiant’ were significantly greater than for ‘UC Keystone’ (ordinal scores increase as disease symptoms increase).

‘UC Keystone’ resistance to *Verticillium* wilt and *Phytophthora* crown rot (PhCR) was also evaluated (see, also Table 3) in comparison to ‘UC Golden Gate’, ‘Monterey’, ‘UCD Royal Royce’, and ‘UCD Valiant’. ‘UC Keystone’ and ‘UC Golden Gate’ are moderately resistant to these diseases. Their resistance phenotypes are not significantly different from those of ‘San Andreas’ and ‘Monterey’, and other commercially important cultivars.

TABLE 2

Across-year estimated-marginal means (EMMs) ^a for <i>Fusarium</i> wilt, <i>Verticillium</i> wilt, and <i>Phytophthora</i> crown rot (PhCR) resistance scores ^b for ‘UC Keystone’, ‘UC Golden Gate’, ‘UCD Valiant’, ‘UCD Royal Royce’, ‘San Andreas’, and ‘Monterey’ observed in 2019-20, 2020-21, and 2021-22 disease resistance screening trials in Davis, CA.					
Disease ^c		Test Cultivar		t ^e	p-value ^f
		Check Cultivar EMM ^d	Check Cultivar EMM		
<i>Fusarium</i> Wilt	UCD Valiant	1.25	2.88	-3.86	0.0003
	UCD Royal Royce		2.98	-4.12	0.0001
	San Andreas		1.15	0.19	0.85
	Monterey		3.36	-4.67	<0.0001
	UC Golden Gate		1.17	0.19	0.85
<i>Verticillium</i> Wilt	UCD Valiant	2.12	1.93	0.43	0.67
	UCD Royal Royce		2.31	-0.44	0.66
	San Andreas		1.29	1.19	0.24
	Monterey		2.36	-0.46	0.64
	UC Golden Gate		1.97	0.34	0.74
<i>Phytophthora</i> Crown Rot	UCD Valiant	2.44	2.25	0.35	0.73
	UCD Royal Royce		2.33	0.21	0.84
	San Andreas		1.83	0.85	0.40
	Monterey		3.07	-0.92	0.36
	UC Golden Gate		2.25	0.34	0.73

^aEMMs were estimated from four replications/entry/year over three years (12 observations/entry), except for *Fusarium*, where additional screening in 2022 resulted in 24 observations/entry.

^bThe ordinal symptom rating scales were identical for each disease: 1 = highly resistant, 2 = moderately resistant, 3 = moderately susceptible, 4 = susceptible, and 5 = highly susceptible.

^cThe fungal pathogens causing these diseases are *Fusarium oxysporum* f. sp. *fragariae* (*Fusarium* wilt), *Verticillium dahliae* (*Verticillium* wilt), *Phytophthora cactorum* (*Phytophthora* crown rot).

^dThe ‘test’ cultivars were ‘UC Keystone’ (shown in the upper half of the table) and ‘UC Golden Gate’ (shown in the lower half of the table).

^et-statistics for linear contrasts (EMM₁-EMM₂) between the EMMs for ‘UC Keystone’ or ‘UC Golden Gate’ and check cultivars.

^fThe probability of a greater t-statistic by chance for tests of the null hypothesis of no difference between EMMs (H₀: EMM₁ = EMM₂).

Photoperiod Insensitive Phenotype

The dominant PF allele is necessary for photoperiod insensitive flower in cultivated strawberry. The predicted photoperiod insensitive phenotype of ‘UC Keystone’ was confirmed through three years of on-farm testing in Santa Maria, California and Prunedale, California. ‘UC Keystone’ consistently flowered and fruited throughout the commercial day-neutral growing seasons in Nipomo (35.0N) and Prunedale (36.8N) where daylengths ranged from 9.8 to 14.7 hours.

Field Evaluations

Three seasons of advanced testing of selected day-neutral hybrids originating from the 2017-18 breeding cycle, start-

ing with ninety-four in 2018-19 and finishing with six in 2021-22, one of which was ‘UC Keystone’, were completed. Comparison cultivars were ‘UC Golden Gate’, ‘Monterey’, ‘UCD Royal Royce’, and ‘UCD Valiant’.

Clones (asexually propagated bare-root plants) for the first year of testing (Phase I in 2018-19) were produced in Winters, California. Clones for subsequent years of testing (Phase II in 2019-20 and 2020-21 and Phase III in 2021-22 and 2022-23) were produced in commercial high-elevation nurseries (Dorris, California) using standard production and propagation practices and post-harvest chilling treatments optimized for the day-neutral market segment. Clones were harvested in late October, trimmed, and directly planted in late October and early November of each year.

For on-farm yield trials, plants were grown in two 10-plant plots in Phase I and two 20-plant plots in Phase II yield trials, and two 500- or 1,000-plant strips in Phase III yield trials. The plots were arranged in randomized complete blocks experiment designs in commercial production fields. These experiments were grown using the management practices, bed configurations, plastic mulches, planting densities, planting dates, irrigation, fertilization, and pesticide application decisions and schedules, and harvest schedules of our cooperators.

Marketable fruit yield, count, and size were recorded at each harvest. Collection and analysis of 18,526 observations were employed for these traits to support statistical analyses.

Fruit Quality

Fruit from early and peak season harvests from every trial were performed to phenotype hybrids for firmness, total soluble solids (TSS=Brix), and titratable acidity (TA). The peak and late season fruit from four environments were stored at 4° C. and screened for post-harvest perishability over 14 days. These samples were phenotyped for weight, TSS, TA, firmness, gloss, and leakage at 0, 7, and 14 days post-harvest (DPH). Over three growing seasons, 2,273 phenotypic observations were collected for fruit quality traits at harvest to support statistical analyses and selection decisions.

Plants were assessed for flavor and aroma through informal hedonic testing in the field and laboratory by various individuals. Although specific volatile organic compounds (VOCs) known to affect aroma were not selected for, selection pressure for enhanced flavor and aroma was applied and VOC profiles were analyzed for 152 cultivars and other hybrids over two years of on-farm testing. ‘UC Keystone’ was among the hybrids phenotyped for VOC. These data were used to assess the effect of subjective olfactory and gustatory assessments on aroma profile changes. VOC analyses were performed using solid phase microextraction (SPME) gas chromatography-mass spectroscopy of fruit samples collected from early and peak season harvests over two years from on-farm yield trials in Santa Maria and Prunedale, California. Although at least 360 VOCs have been identified in developing strawberry receptacles, aroma and flavor are dominated by fewer than 10. Data for approximately 52 VOCs affecting aroma (58,488 phenotypic observations) were collected and analyzed to support statistical analyses. Data for three VOCs (γ-decalactone, mesifurane, and linalool) that are predicted to contribute towards the improved flavor of ‘UC Keystone’ and ‘UC Golden Gate’ are provided.

The fruit quality of ‘UC Keystone’ equal or exceeds the fruit quality and shelf life standards of long shelf life (LSL) cultivars designed for prolonged cold storage and long-distance shipping. Table 3 provides a comparison of the cumulative marketable fruit yields and other fruit traits.

TABLE 3

Across-environment estimated marginal means (EMMs) for cumulative marketable fruit yield and other fruit traits for ‘UC Keystone’ and comparison cultivars grown on commercial farms in Nipomo and Prunedale, CA over three growing seasons (2019-20 to 2021-22)^a.

Trait ^a	Comparison Cultivar	UC Keystone EMM	Comparison Cultivar EMM	t ^b	p-value ^c
Yield (lb/acre)	Monterey	103,710	88,323	1.42	0.16
	UCD Valiant		91,783	1.10	0.28
	UCD Royal		89,044	1.35	0.18
	Royce				
	UC Golden Gate		95,730	0.73	0.47
Size (g/fruit)	Monterey	28.0	29.7	-3.23	0.001
	UCD Valiant		33.0	-8.91	<0.0001
	UCD Royal		28.8	-1.39	0.16
	Royce				
	UC Golden Gate		28.7	-1.20	0.23
Firmness (g-force)	Monterey	260.6	239.1	1.96	0.05
	UCD Valiant		297.1	-2.93	0.005
	UCD Royal		349.9	-7.18	<0.0001
	Royce				
	UC Golden Gate		303.7	-3.25	0.002
TSS (%)	Monterey	8.29	8.89	-4.19	0.0001
	UCD Valiant		7.37	5.78	<0.0001
	UCD Royal		7.57	4.55	<0.0001
	Royce				
	UC Golden Gate		7.84	2.59	0.01
TA (%)	Monterey	0.77	0.76	-0.39	0.70
	UCD Valiant		0.82	-2.97	0.005
	UCD Royal		0.82	-2.87	0.006
	Royce				
	UC Golden Gate		0.85	4.21	0.0001
TSS/TA	Monterey	11.01	10.44	-2.69	0.01
	UCD Valiant		9.20	6.35	<0.0001
	UCD Royal		9.55	5.12	<0.0001
	Royce				
	UC Golden Gate		9.45	5.19	<0.0001

^aCumulative marketable fruit yields were estimated from fruit harvested on commercial schedules (once or twice weekly) over the entire day-neutral growing season on each farm. Fruit firmness, total soluble solids (TSS), and titratable acidity (TA) were measured from multiple fruit/replication sampled from two harvests/trial. EMMs and test statistics were estimated from the phenotypes of fruit harvested from two 24-plant plots/entry/environment.

^bt-statistics for linear contrasts (EMM₁-EMM₂) between ‘UC Keystone’ and check cultivar EMMs.

^cThe probability of a greater t-statistic by chance for tests of the null hypothesis of no difference between EMMs (H₀: EMM₁ = EMM₂).

‘UC Keystone’ produced large fruit (28.0 to 28.7 g/fruit) in the on-farm trials (Table 3). Their fruit weights were in the normal range for mass-produced day-neutral cultivars in California and above the market critical threshold needed for cost-effective hand harvest (labor costs increase as fruit size decreases).

‘UC Golden Gate’ fruit were significantly firmer (303.7 g-force) than ‘UC Keystone’ (260.6 g-force; p=0.0002) and ‘Monterey’ (239.1 g-force; p<0.0001). The firmness of these cultivars exceeded the threshold (>230 g-force) necessary for minimizing harvest and post-harvest losses and delivering a high quality product to markets over long distances.

The sweetness and sourness profiles of ‘UC Keystone’ and ‘UC Golden Gate’ are slightly different. Total soluble

solids (TSS) were slightly greater for ‘UC Keystone’ (8.33) than ‘UC Golden Gate’ (7.85; p=0.008), whereas titratable acidity (TA) was greater for ‘UC Golden Gate’ (0.83) than ‘UC Keystone’ (0.74; p=0.0003). The TSS/TA ratio (a measure of perceived sweetness) was significantly greater for ‘UC Keystone’ (11.66) than ‘UC Golden Gate’ (9.86; p=0.0001) and non-significantly greater than ‘Monterey’ (10.44; p=0.01), whereas the TSS/TA ratio for ‘UC Golden Gate’ was significantly lower than ‘Monterey’ (p<0.0001). Hence, the sweetness and perceived sweetness ratings for ‘UC Keystone’ and ‘Monterey’ were virtually identical and greater than that of ‘UC Golden Gate’.

‘UC Keystone’ and ‘UC Golden Gate’ were identified to have improved aroma from subjective olfactory and gustatory assessments of fruit. Solid phase microextraction (SPME) GC-MS analyses of volatile organic compounds (Table 4) suggested that part of the improved flavor of ‘UC Keystone’ and ‘UC Golden Gate’ can be attributed to an increase in γ -decalactone (5-hexyloxolan-2-one), the most abundant lactone found in strawberry. This VOC, which imparts an intense-peach flavor, is abundant in ‘Albion’ (U.S. Plant Pat. No. 16,228) and absent in ‘Monterey’.

TABLE 4

Across-year estimated-marginal means (EMMs)^a for γ -decalactone, mesifurane, and linalool concentrations^a in ripe fruit of ‘UC Keystone’, ‘UC Golden Gate’, and comparison cultivars harvested in 2020-21 and 2021-22 from on-farm yield trials in Nipomo, CA.

Cultivar	γ -decalactone EMM	mesifurane EMM	linalool EMM
Monterey	0	1,025,275	58,682
UCD Royal Royce	7,253	247,148	0
UCD Valiant	14,406	213,015	1,374
UC Keystone	34,740	156,584	20,783
UC Golden Gate	7,273	241,300	6,265
Albion	132,592	991,288	17,942
LSD ^b	109,763	146,678	26,860

^aEMMs were estimated from fruit sampled from two 20- or 40-plant plots/entry/year. The relative concentrations of volatile organic compounds (ng/g dry weight) were estimated from the raw solid phase microextraction (SPME) GC-MS data (peak area in counts) and normalized samples.

^bLeast significant differences are shown for a false-positive probability of p = 0.05.

‘UC Keystone’ and ‘UC Golden Gate’ meet or exceed the shelf life requirements for mass-produced cultivars developed for cold storage and long distance shipping in California. Minimal declines were observed in the deterioration of fruit firmness and other fruit quality traits over 14 days of post-harvest cold storage using fruit sampled from peak and late season harvests. No statistically significant differences for fruit weight, TSS, TA, or leakage between ‘UC Keystone’ and ‘UC Golden Gate’ and comparison cultivars were observed.

Fruit Production Characteristics—Field Trials

The number of harvests ranged from 40 to 61 in each trial to assess ‘UC Keystone’ fruit production in comparison to that of ‘UC Golden Gate’, ‘Monterey’, ‘UCD Royal Royce’, and ‘UCD Valiant’. Marketable fruit yield, count, and size were recorded at each harvest. Collection and analysis of 18,526 observations was performed for these traits to support statistical analyses.

Statistics are shown for linear contrasts between the estimated marginal means (EMMs) for ‘UC Keystone’, ‘UC Golden Gate’, and comparison cultivars across environment (2 locations 3 years) and for individual environments to highlight variation in planting dates, the number of harvests, harvest ranges, and production practices (Table 5). EMMs were estimated from two 20-plant plots (replications)/entry/environment in 2020 and 2021 and two 540-plant plots in 2022.

Location	Season	Comparison Cultivar	UC Keystone EMM (lb/a)	Compare Cultivar EMM (lb/a)
Nipomo	2019-20	Monterey	88,028	64,797
		UC Royal Royce		79,625
		UC Valiant		90,853
		UC Golden Gate		82,971
Prunedale	2019-20	Monterey	103,149	73,300
		UC Royal Royce		101,146
		UC Valiant		81,841
		UC Golden Gate		103,065
Nipomo	2020-21	Monterey	99,214	70,682
		UC Royal Royce		93,142
		UC Valiant		107,279
		UC Golden Gate		99,935
Prunedale	2020-21	Monterey	115,908	97,546
		UC Royal Royce		128,842
		UC Valiant		119,684
		UC Golden Gate		123,669
Nipomo	2021-22	Monterey	56,691	69,976
		UC Royal Royce		39,241
		UC Valiant		57,602
		UC Golden Gate		59,740
Prunedale	2021-22	Monterey	158,914	152,127
		UC Royal Royce		92,924
		UC Valiant		93,437
		UC Golden Gate		104,999

Location	Yield Change (%) ^b	t ^c	p-value ^d
Nipomo	35.9	3.70	0.004
	10.6	1.03	0.33
	-3.1	-0.45	0.66
	6.1	0.84	0.42
Prunedale	40.7	5.56	0.0002
	2.0	0.37	0.72
	26.0	3.97	0.002
	0.1	0.02	0.99
Nipomo	40.4	3.16	0.01
	6.5	0.67	0.52
	-7.5	-0.89	0.39
	-0.7	-0.08	0.94
Prunedale	18.8	2.92	0.01
	-10.0	-2.05	0.06
	-3.2	-0.60	0.56
	-6.3	-1.23	0.24
Nipomo	-19.0	-5.17	<0.0001
	44.5	5.54	<0.0001
	-1.6	-0.29	0.78
	-5.1	-0.97	0.34

-continued

Prunedale	4.5	1.25	0.23
	71.0	9.89	<0.0001
	9.8	70.1	<0.0001
	51.3	8.08	<0.0001

^aLinear contrasts between estimated marginal means (EMMs) for ‘UC Keystone’ and comparison cultivars were estimated for each environment. EMMs were estimated from two 20-or 40-plant plots/entry within each environment. Cumulative marketable fruit yields were estimated from fruit harvested on commercial schedules (once or twice weekly) over the entire day-neutral growing season on each farm.

^bThe yield change percentage was estimated by $(EMM_1 - EMM_2)/EMM_2 \times 100$.

^ct-statistics for linear contrasts $(EMM_1 - EMM_2)$ between the EMMs for ‘UC Keystone’ and comparison cultivars.

^dThe probability of a greater t-statistic by chance for tests of the null hypothesis of no difference between EMMs ($H_0: EMM_1 = EMM_2$).

‘UC Keystone’ and ‘UC Golden Gate’ were the highest yielding cultivars in these trials. When averaged across environments, however, the yields of ‘UC Keystone’, ‘UC Golden Gate’, and the three comparison cultivars were not statistically different. Thus, ‘UC Keystone’ has yields at least equal to the *Fusarium* wilt-susceptible comparison cultivars.

The cumulative marketable fruit yields of ‘UC Keystone’ were 6,787 to 29,849 lb/a greater than ‘Monterey’ in five environments and 13,285 lb/a less than ‘Monterey’ in one environment (Nipomo 2021-22) (Table 5).

The cumulative marketable yield plots for each environment highlight the differences among cultivars and growing seasons. The yields of ‘Monterey’ were below and separated from ‘UC Keystone’, ‘UC Golden Gate’, and the other comparison cultivars in the first two growing seasons. Yields dropped in the last growing season (2021-22) in Nipomo especially compared to previous years, with less separation among cultivars.

The cumulative marketable fruit yields of ‘UC Keystone’ and ‘UC Golden Gate’ were not significantly different in five of six test environments or across environments (Tables 3 and 5). The yield of ‘UC Keystone’ was significantly greater than ‘UC Golden Gate’ in one environment (Prunedale 2021-2022). The reasons for yield decreases for ‘UC Golden Gate’, ‘UCD Royal Royce’, and ‘UCD Valiant’ in the 2021-22 Prunedale trial were not clear. The affected plants ultimately recovered, however, and produced an abundance of fruit later in the season.

‘UC Keystone’ is thus a highly productive *Fusarium* wilt resistant cultivar for the day-neutral segment that meets or exceeds the shelf life, flavor, and fruit quality characteristics of ‘Monterey’.

What is claimed is:

1. A new and distinct cultivar of strawberry plant named ‘UC Keystone’, substantially as described and illustrated herein.

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FIG. 1



FIG. 2